Digital Games in Museums: Constructing a Framework of playfulness

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When you enter a museum as a visitor, your experience raises important questions for museums: Are visitors having memorable and enjoyable experiences? Do the exhibitions effectively provide knowledge or fun? Are visitors fully engaged with the exhibits? Among these considerations, one concept emerges as a potential solution: playfulness. In museums, playfulness enriches engagement, contributes to visitor wellbeing, and increases repeat visits (Derry, 2021). It usually manifests through playful interactions with digital installations such as games (Vosinakis et al. 2020), interactive artworks (Ryding & Fritsch, 2020), and mobile applications (Hsu et al., 2018).

Building on ludic studies and museum games research, this project aims to construct the first conceptual framework of 'playfulness' in the museum context by exploring playful visitor experiences with interactive exhibits.

Playfulness is a compound concept examined from multiple perspectives. It can be viewed as relating to the person, such as a personality characteristic (Holopainen & Stain, 2015), a lifelong attitude (Bauman, 1995), or a mindset for approaching life (Lucero et al., 2014). Alternatively, playfulness can be understood as a feature of an environment or artifact (Holopainen & Stain, 2015), or as a design quality for products that goes beyond mere entertainment (Lucero et al., 2014). Masek and Stenros (2021) identify 'clusters of meaning' of playfulness in the literature that emphasise 'engagement' over external consequence, realness, or convention, and help organise the environment for engagement. However, there is a lack of systematic exploration into the broader structures of playfulness in the museum, or the ways in which exhibit design can structure playful and meaningful engagement. Understanding these aspects is important for designing better exhibits and for determining how playfulness can be harnessed to create engaging visitor experiences.

Focusing on the playful states of mind when museum visitors interact with exhibits and the playful affordances of exhibits, this project asks three research questions:

1. What characterises a playful experience, its context, and outcomes in the museum?
2. Are some interactive exhibits more effective at prompting play and evoking playful experiences in visitors? If so, what are their common characteristics and structures and how do they structure playful engagement?
3. How do playful designs help create better visitor experiences?

The first two questions focus on the playful experiences of visitors when they engage with interactive exhibits in museums. Findings related to these questions will contribute to identifying the common characteristics of playful experiences and to developing a conceptual framework of playfulness in the museum. Finally, this research will offer implications and suggestions for improving playful designs to enhance visitor experiences in museums.

This research follows a multi-phase, multi-method approach, aiming to build a conceptual framework for playfulness in the museum context by examining three aspects: playfulness as a psychological trait, visitors' playful experiences engaging with selected exhibits, and the design affordances of exhibits for playfulness. Birmingham Thinktank Science Museum was selected as the case study site in the UK. Linfen Museum in Shanxi Province, China, served as an additional site, chosen to increase the sample size and to provide a comparative cultural perspective. After three rounds of expert filtering and evaluation, two interactive exhibits at Thinktank and three at Linfen Museum were determined to be appropriate for data collection, offering diverse physical and digital interactions.

Based on current studies on playfulness, game experience, gamification, and playful experience models or frameworks, a visitor questionnaire was designed to collect data on playfulness as a personality trait, playful experiences, and demographic data such as gender and age range. Fieldwork in the UK and China was successfully conducted, and 239 valid samples were collected. Additionally, two interviews with museum experts were conducted, gathering their ideas on play and playful design in museums.

The collected data have been organised and are currently in the analysis stage. To identify the characteristics of playfulness as both a state of mind and a feature of exhibits, four sections of data analysis will be conducted, primarily based on the questionnaire design.

The initial section of the analysis focuses on the Adult Playfulness Trait Scale (Shen et al., 2014a; 2014b). The Adult Playfulness Trait Scale (APTS) is used to evaluate playfulness as a psychological personality trait and to examine whether individuals higher in playfulness tend to experience more playful interactions when interacting with museum exhibits. Within this section, the Chi-Square test and Ordinal Logistic Regression will be employed to examine the correlation between this trait and visitors' playful experience.

Following this, the analysis will examine the Playful Experience Scale. Visitors' playful experiences when interacting with the five selected exhibits are essential to this research, and results from this scale reflect two aspects of playfulness: the mindful state of visitors and the exhibit affordance of interactive exhibits. To address the mindful state aspect, the Cronbach's Alpha test and Confirmatory Factor Analysis will be used to assess the scale's reliability and validity, determining whether the visitor data aligns with established theoretical structures. To understand exhibit affordance, descriptive analysis of the scale data will be combined with expert evaluations and interviews regarding the exhibits to help identify whether specific design elements are particularly effective in fostering playful experiences among visitors.

Further analysis will encompass the remaining data from the questionnaire. Here, qualitative and quantitative methods will be applied to the demographic data to explore potential relationships and investigate whether these factors offer possibilities for extending the playfulness framework. This exploration includes examining potential correlations between demographic variables and the playfulness trait or playful experience; differences in playful experience levels based on gender or exhibit; variations in playfulness across age groups; and the influence of co-visitors on playful experience, among other factors.

The final stage of the analysis involves the construction of the final framework. This final framework of playfulness in the museum context will synthesize all results from the previously mentioned approaches, including outcomes from the APTS, the Playful Experience Scale, the remaining questionnaire data analysis, expert evaluation, and expert interviews. Based on these synthesized outcomes, the research will identify common structures of playful experience and develop a conceptual framework of playfulness within the museum context. This framework is designed to encompass three perspectives on the concept: personality trait, mindful state, and exhibit affordance. Furthermore, the overall concept of playfulness will be formulated specifically for the museum setting, and differences in playfulness between the UK and China will also be discussed.

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